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DUAL THRUSTER ASSEMBLY FOR MARITIME VEHICLE

Abstract

A jet pump assembly for causing a maritime vehicle to traverse a body of water, and a maritime vehicle including such a jet pump assembly. The jet pump assembly includes a jet pump drive system including an impeller and a motor operatively coupled to the impeller. The jet pump assembly also includes a transom assembly that includes a transom plate including a first aperture facing the motor, a transom housing coupled to the transom plate, thereby enclosing the first aperture, and a transom box coupled to the transom plate and including a second aperture aligned with the first aperture. The jet pump assembly further includes a bottom plate coupled to and disposed perpendicular to the transom plate. The bottom plate includes a base and an opening formed through the base, wherein the opening is enclosed by the transom housing. The impeller is at least partially disposed within the transom box.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/561,166, titled “Jet Pump Assembly for Maritime Vehicle” and filed on Mar. 4, 2024, the contents of which are hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to jet pump systems and, more particularly to a dual thruster assembly for a maritime vehicle.

BACKGROUND

[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.

[0004] Maritime vehicles generally utilize thrust systems for causing movement of the maritime vehicle. In some cases, maritime vehicles utilize propeller-based thrust systems. Maritime vehicles can alternatively utilize jet-based pump thrust systems, whereby propellers extend outside the hull of the maritime vehicle while jet pumps pass water through a channel and through an impeller disposed within the maritime vehicle to generate thrust. The jet pump forces the water through a directional nozzle to control the direction of thrust. As a result, the jet pump controls the forward speed and direction of the maritime vehicle.

[0005] Additionally, maritime vehicles require various cooling systems to expel the heat generated in operating the vehicle. For example, thrust systems generally include internal combustion or electrical motors that generate large amounts of thermal energy. Cooling systems absorb heat from the thrust system and transfer the thermal energy into the surrounding environment. In some examples, maritime vehicles can incorporate heat exchangers that direct the thermal energy into the surrounding water.

SUMMARY

[0006] A jet pump assembly for causing a maritime vehicle to traverse a body of water, and a maritime vehicle including such a jet pump assembly. The jet pump assembly includes a jet pump drive system including an impeller and a motor operatively coupled to the impeller. The jet pump assembly also includes a transom assembly that includes a transom plate including a first aperture facing the motor, a transom housing coupled to the transom plate, thereby enclosing the first aperture, and a transom box coupled to the transom plate and including a second aperture aligned with the first aperture. The jet pump assembly further includes a bottom plate coupled to and disposed perpendicular to the transom plate. The bottom plate includes a base and an opening formed through the base, wherein the opening is enclosed by the transom housing. The impeller is at least partially disposed within the transom box.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] The present disclosure is described in the following detailed description in conjunction with the drawings, wherein:

[0008] FIG. **1**A is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure.
[0009] FIG. **1**B is a front view of the maritime vehicle of FIG. **1**A.
[0010] FIG. **1**C is a rear view of the maritime vehicle of FIG. **1**A.
[0011] FIG. **1**D is a bottom perspective view of the maritime vehicle of FIG. **1**A.
[0012] FIG. **1**E is another bottom perspective view of the maritime vehicle of FIG. **1**A.
[0013] FIG. **1**F is similar to FIG. **1**A, but with the cap and various components of the maritime vehicle removed for illustrative purposes.
[0014] FIG. **1**G is similar to FIG. **1**F, but with additional components of the maritime vehicle removed for illustrative purposes.
[0015] FIG. **1**H is a rear view of the maritime vehicle of FIG. **1**F.
[0016] FIG. **1**I is a first cross-sectional view taken along line I-I in FIG. **1**A.
[0017] FIG. **1**J is a second cross-sectional view taken along line J-J in FIG. **1**A.
[0018] FIG. **1**K is similar to FIG. **1**A, but with the latching assembly of the maritime vehicle removed for illustrative purposes.
[0019] FIG. **2** is a front perspective view of an example dual jet pump assembly made in accordance with the present disclosure and usable in connection with the maritime vehicle of FIGS. **1**A-**1**K.
[0020] FIG. **3** is a perspective view of the jet pump drive systems of the jet pump assembly of FIG. **2**.
[0021] FIG. **4** is a perspective view of the motors of the jet pump drive systems of FIGS. **2** and **3**.
[0022] FIG. **5** is a perspective view of the drive shaft and impeller of one of the jet pump drive systems of FIGS. **2** and **3**.
[0023] FIG. **6** is a perspective view of the jet pump nozzle of one of the jet pump drive systems of the jet pump assembly of FIG. **2**.
[0024] FIG. **7** is a perspective view of the transom assembly of the jet pump assembly of FIG. **2**.
[0025] FIG. **8** is a perspective view of one of the transom housings of the transom assembly of FIG. **7**.
[0026] FIG. **9** is a side view of the transom housing of FIG. **8**.
[0027] FIG. **10**A is a front, perspective view of the transom plate of the transom assembly of FIG. **7**.
[0028] FIG. **10**B is a rear plan view of the transom plate of FIG. **10**A.
[0029] FIG. **11** is a front, perspective view of the transom box of the transom assembly of FIG. **7** and portions of the jet pump drive systems of FIGS. **2** and **3**.
[0030] FIG. **12** is a rear perspective view of FIG. **2**.
[0031] FIG. **13**A is a front perspective view of the bottom plate of the dual jet pump assembly of FIG. **2**.
[0032] FIG. **13**B is a rear perspective view of FIG. **13**A.
[0033] FIG. **14** is a perspective bottom view of the bottom plate of the dual jet pump assembly of FIG. **2**.
[0034] FIG. **15** is a perspective view of the jet pump assembly of FIG. **2** installed in the maritime vehicle of FIGS. **1**A-**1**K.
[0035] FIG. **16** is a bottom view of the jet pump assembly of FIG. **2** installed in the maritime vehicle of FIGS. **1**A-**1**K.
[0036] FIG. **17** is a schematic diagram of one example of a cooling system made in accordance with the present disclosure and usable to cool the jet pump assembly of FIG. **2**.
[0037] FIG. **18**A is a perspective top view of one example of a micro-keel cooler made in accordance with the present disclosure and usable in the cooling system of FIG. **17**.
[0038] FIG. **18**B another perspective top view of the micro-keel cooler of FIG. **18**A but with an end plate removed.

[0039] FIG. **19** is a perspective bottom view of the micro-keel cooler of FIGS. **18**A and **18**B.
[0040] FIG. **20** is a perspective view of a bottom portion of the micro-keel cooler of FIGS. **18**A and **18**B.
[0041] FIG. **21** is a perspective view of a top portion of the micro-keel cooler of FIGS. **18**A and **18**B.
[0042] FIG. **22** is a perspective view of one of the end plates of the micro-keel cooler of FIGS. **18**A and **18**B.
[0043] FIG. **23** is a close-up, bottom perspective view of a portion of the maritime vehicle of FIGS. **1**A-**1**K and including two of the micro-keel coolers of FIGS. **18**A and **18**B.
[0044] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments. It will further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

DETAILED DESCRIPTION

[0045] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is small(er), durable, and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.
[0046] FIGS. **1**A-**1**K illustrate one example of a maritime vehicle **100** constructed in accordance with the teachings of the present disclosure. The maritime vehicle **100** is an unmanned vessel configured to autonomously traverse a body of water. The maritime vehicle **100** generally includes a hull **104** and a cap **108** that is coupled to the hull **104** to secure various components within the maritime vehicle **100**. The hull **104** is at least partially disposed in the body of water in which the maritime vehicle **100** is traversing. The hull **104** in this example is a mono-hull that has a front (or bow) **112**, a rear (or stern) **116**, two sides **120**, and a keel **124** coupled to another. The front **112**, the rear **116**, the sides **120**, and the keel **124** can be welded together or can be coupled to one another in a different manner. For example, the front **112**, the rear **116**, the sides **120**, and the keel **124** can be coupled together in the manner described in U.S. Provisional Application No. 63/561,282, titled "Systems and Approaches for Assembling a Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The hull **104** is configured such that the hull provides a continuous planning surface that allows the maritime vehicle **100** to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap **108** is coupled to the hull **104** to cover and/or conceal the components of the maritime vehicle **100** disposed in and carried by the hull **104** as the maritime vehicle **100** traverses the body of water.
[0047] In this example, the hull **104** and the cap **108** each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can

be equal to approximately 14 feet. The hull **104** is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle **100** can include two or more hulls (e.g., two parallel hulls) instead of the mono-hull. In this example, the cap **108** entirely covers the hull **104** (and the components therein). In other examples, however, the maritime vehicle **100** need not include the cap **108** or the cap **108** may only partially cover the hull **104** (and the components disposed therein).

[0048] In some examples, the cap **108** can be removably coupled to the hull **104** via a locking system. For example, as illustrated in FIGS. **1**A-**1**K, the locking system **128** can take the form of a plurality of latch mechanisms **128** disposed around a perimeter of the maritime vehicle **100**. Thus, the cap **108** can be removed to allow access to the interior of the hull **104**. In other examples, however, the cap **108** can be permanently coupled to the hull **104** to permanently conceal the components within the maritime vehicle **100**.

[0049] The maritime vehicle **100** also includes a plurality of bulkheads **132** arranged within the hull **104**. The bulkheads **132** divide the maritime vehicle **100** into a plurality of different compartments for receiving and retaining different components in the maritime vehicle **100**.

[0050] The maritime vehicle **100** also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle **100** as well as data about the environment surrounding the maritime vehicle **100** (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle **100**. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle **100**, the pressure of the water in which the maritime vehicle **100** is disposed, the pressure within the maritime vehicle **100**), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle **100**, a temperature of ambient air external to the maritime vehicle **100**, a temperature of water in which the maritime vehicle **100** is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water sensors (e.g., a float switch, a capacitive sensor, an ultrasonic sensor, an electrical water sensor, etc.) to determine when water is present and/or present to a given extent (e.g., at a certain volume or level), one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof.

[0051] The maritime vehicle **100** also includes a power system that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power system generally includes a thrust system and one or more power sources configured to power the thrust system (and the other components within the maritime vehicle **100**). The thrust system is generally configured to propel the maritime vehicle **100** in/on/along the water. The thrust system can be a propeller-based thrust system or can be a jet pump-based thrust system. The one or more power sources can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the maritime vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. **1**A-**1**K includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** illustrated in FIGS. **2**A-**2**K also includes a retention assembly for the four battery assemblies, e.g., the retention assembly described in U.S. Provisional Application No. 63/561,063, titled “Power System for Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The maritime vehicle **100** generally also includes a cooling system configured to cool the thrust system and/or the one or more power sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle **100**.

[0052] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment. Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0053] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, for example a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and re-surfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its desired location in the body of water (or in a different body of water). The vision system is generally configured to capture, process, and analyze images obtained by one or more image sensors and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment).

[0054] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or more local controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae disposed on an exterior of the cap **108** as well as a plurality of antennae disposed in the hull **104**.

[0055] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more thrust controllers (for controlling the operation of the thrust system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0056] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the maritime vehicle **100** (or between the maritime vehicle **100** and the environment surrounding the maritime vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters, one or more supervisor boards, one or more brain boards.

[0057] Further described herein is a compact and lightweight jet pump assembly and micro-keel cooler system for a small maritime vehicle (e.g., the maritime vehicle **100**). One example of such a compact and lightweight jet pump assembly **200** is illustrated in FIGS. **2-16**. In this example, the jet pump assembly includes two jet pumps in a single assembly. In other examples, the jet pump assembly can include one jet pump or more than two jet pumps. Additionally, each of the jet pumps is preferably coupled to a cooling system such as the cooling system described in connection with FIG. **17** or the micro-keel cooler system, one example of which is illustrated in and described in connection with FIGS. **18-22**. In some examples, a single micro-keel cooler system can be coupled to two or more jet pumps.

Jet Pump Assembly

[0058] Jet pump assemblies generally include a water intake, an impeller, a diffuser, a nozzle, and a steering system. Overall, jet pump assemblies offer a compact and efficient means of propulsion and steering for small maritime vehicles (e.g., small boats and small vessels) and jet skis, making them well-suited for use in shallow water and areas where traditional propellers may be impractical or unsafe.

[0059] The dual jet pump assembly **200** illustrated in FIGS. **2-16** is made in accordance with the present disclosure and is configured for use in a maritime vehicle such as the maritime vehicle **100**. As illustrated in FIG. **2**, the dual jet pump assembly **200** includes two jet pump drive systems **202** (described in greater detail in connection with FIGS. **3-5**), a transom assembly **204** (described in greater detail in connection with FIGS. **7-12**), and a bottom plate **206** (described in greater detail below in connection with FIGS. **13**A, **13**B, and **14**). The dual jet pump assembly **200** is manufactured as a functional unit that can be assembled prior to installation in the maritime vehicle (and easily and quickly removed from the maritime vehicle when replacement or repair is necessary). For example, the dual jet pump assembly **200** can be manufactured and bench tested prior to installation in the maritime vehicle. While in the present example, the dual jet pump assembly **200** includes two jet pump drive systems **202** and the transom assembly **204**, configured for the two jet pump drive systems **202**, in other examples, the jet pump assembly can include one jet pump drive system or more than two jet pump drive systems.

Jet Pump Drive System

[0060] FIGS. **3-6** illustrate the jet pump drive systems **202** of FIG. **2** in greater detail. As illustrated, each jet pump drive system **202** includes a motor **302** (shown in detail in FIGS. **3** and **4**), a drive shaft **306** (shown in detail in FIGS. **3** and **5**), an impeller **308** (shown in detail in FIGS. **3** and **5**), and a nozzle **602** (shown in FIG. **6**). Each jet pump drive system **202** is identical to the other, but in some examples, the jet pump drive systems **202** can be mirrored or otherwise altered to meet the functional requirements of the maritime vehicle (e.g., maritime vehicle **100**). In the present example, the motor **302** is coupled to the drive shaft **306** via a coupling **312** and the impeller **308** is coupled to (e.g., integrally formed with) the drive shaft **306**. In other examples, the drive shaft **306** can be integrally formed with the motor **302** and/or the impeller **308** can be coupled to the motor **302** via a coupling similar to the coupling **312**. Each component of the jet pump drive system **202** will be described in greater detail below.

[0061] The motors **302** are electric motors and can have any functional number of phases or poles.

Each motor **302** includes various electrical power and control connections **402**. Each motor **302** may include a motor controller (e.g., an electronic speed controller (ESC)), not shown, to control the speed of the motor **302**. The motor controller can be integrated in the motor **302** or can be disposed separate from and in electric communication with the motor **302**. In other examples, however, the motors **302** can be any type of motor, engine, or power source (e.g., internal combustion motor). Additionally, because in the present example each jet pump drive system **202** includes one motor **302** for each drive shaft **306** and impeller **308**, the two motor controllers can independently control the two motors **302** (and the two drive shafts **306** and impeller **308** respectively coupled thereto), respectively. In other examples, however, a single motor **302** can be configured to actuate both drive shafts **306**. In such examples, the single motor **302** can be coupled to the drive shafts **306** via a gear assembly.

[0062] During operation, each motor **302** generates excess thermal energy and can get quite hot. Accordingly, the motors **302** often require cooling. As shown in FIG. **4**, each motor **302** in this example includes a cooling jacket **412** and couplings **422**, **424** that couple the cooling jacket **412** to a cooling system such as the cooling system described in greater detail in connection with FIG. **17**. In other examples, the motor **302** may be air cooled or temperature regulated using another cooling method. In such examples, the motor **302** may not include the cooling jacket **412** and/or the couplings **422**, **424**.

[0063] As shown in FIG. **5**, the drive shaft **306** includes a first end **502** and a second end **504**. The first end **502** couples to the motor **302** via the coupling **312** and the second end **504** couples to the impeller **308**. The drive shaft **306** is configured to transmit rotational energy from the motor **302** to the impeller **308**. As a result, the drive shaft **306** is rigid and capable of withstanding the torque of the motor **302**. In various examples, the drive shaft **306** can be made of a strong metal, polymer, or similar material. The drive shaft **306** is cylindrical, but in other examples, the drive shaft **306** can have any other cross-sectional shape that is radially symmetrical and capable of rotating at high revolutions per minute.

[0064] The drive shaft **306** also includes a slot **512**, near the first end **502**, for facilitating the connection with the coupling **312**. The slot **512** may engage with a set screw of the coupling **312** or include a shoulder to releasably couple the coupling **312** to the slot **512**. Additionally, in the present example, the drive shaft **306** is coupled to the impeller **308** at the second end **504** via a coupling **514**. In another example, the second end **504** of the drive shaft **306** can be inserted into an opening of the impeller **308** and welded to the impeller **308**. In yet other examples, the drive shaft **306** can be coupled to the impeller **308** via a different but known method of coupling rotating components. Likewise, the drive shaft **306** can be coupled to the motor **302** in any number of different known manners. For example, the drive shaft **306** can be welded to the motor **302**.

[0065] As also shown in FIG. **5**, the impeller **308** includes impeller blades **552** disposed on an impeller head **554**. The impeller head **554** includes a bearing **556** for coupling to the nozzle **602** (discussed in greater detail in connection with FIG. **6**). In this example, the impeller blades **552** are symmetrically disposed about the head **554** to ensure stability during operation. During operation, the impeller head **554** and the impeller blades **552** are rotated about a central axis **560** and impart momentum to a fluid (e.g., water) in the direction **562**. As a result, the impeller **308** increases the speed of the fluid in the direction **562**.

[0066] The blades **552** of the impeller **308** are curved and shaped to improve the effectiveness of the impeller **308**. In the present example, the impeller **308** includes four blades **552**, but in other examples, the impeller **308** can include more or fewer blades **552**. Additionally, a radial length **564** of the blades **552** can be selected so as to efficiently operate in the fluid. For example, the radial length **564** of the blades **552** can be limited to reduce the possibility of cavitation in water.

[0067] FIG. **6** illustrates one of the nozzles **602** that may couple to the impellers **308**. Each nozzle **602** generally includes a diffuser **604**, a nozzle **606**, and a directional outlet **608**. The diffuser **604** gradually transitions the flow area of the nozzle **606** to create a reactive force to generate a propulsive force. The diffuser **604** includes a tail cone **612** including an aperture **614** sized to receive the bearing **556**. The directional outlet **608** of the nozzle **602** is configured to direct the propulsive force of the fluid exiting the nozzle **602**, thereby permitting control of the direction of movement of the maritime vehicle (e.g., the maritime vehicle **100**). In the present example, the directional outlet **608** is pivotable about the pivoting axis **622**. In other examples, the directional outlet **608** can be actuatable about a different axis or two axes.

Transom Assembly

[0068] The transom assembly **204** of the jet pump assembly **200** provides an enclosed space that receives water from the bottom of the maritime vehicle **100** and, through operation of the jet pump drive systems **202** of FIGS. **2-6**, expels high-speed water from the aft of the vehicle to propel the vehicle forward. The transom assembly **204** generally includes two transom housings **702** (one for each jet pump drive system **202** and described in greater detail in connection with FIGS. **8** and **9**), two transom plates **704** (described in greater detail in connection with FIGS. **10** and **11**), and a transom box **706** (described in greater detail in connection with FIG. **12**). The transom assembly **204** is configured to be coupled with the jet pump drive systems **202** via apertures **712**, respectively. For example, the drive shafts **306** pass through the apertures **712**, as best illustrated in FIG. **2**. In some examples, water-tight seals are utilized to keep the apertures **712** watertight.

[0069] As shown in FIGS. **8** and **9**, each transom housing **702** includes a bottom coupling **802**, an aft coupling **804**, and a transom channel **806** disposed between the bottom coupling **802** and the aft coupling **804**. The bottom coupling **802** of each transom housing **702** is configured to engage the bottom plate **206** as shown in FIG. **2** and described in greater detail in connection with FIGS. **13**A & **13**B. Meanwhile, the aft coupling **804** of each transom housing **702** is configured to engage the transom plate **704** as shown in FIGS. **2** and **7**. The transom channel **806** is disposed and extends between the bottom coupling **802** and the aft coupling **804** and serves as the pump intake for each respective jet pump drive system **202**.

[0070] As shown in FIG. **9**, the bottom coupling **802** of each transom housing **702** includes a coupling plate **822** and a lower plate **824**. The coupling plate **822** is provided to fasten the respective transom housing **702** to the bottom plate **206**. In some examples, the coupling plate **822** can be bolted, riveted, or welded to the bottom plate **206**. The lower plate **824** is configured to be flush with the exterior of a hull (e.g., the hull **104**) of the maritime vehicle when the transom housing **702** and bottom plate **206** are installed on the vehicle. In some examples, the lower plate **824** can further include a protective cover or grate to limit debris entering the transom housing **702**.

[0071] Further, the aft coupling **804** of each transom housing **702** includes a ring **832** and a flange **834**. The ring **832** passes through a respective aperture in the transom plate **704** (described in greater detail below in connection with FIGS. **10** and **11**). In some examples, the nozzles **602** are coupled to the rings **832**, respectively. The rings **832** pass through the transom plate **704** but the flange **834** of each transom channel **702** is coupled to the transom plate **704** via rivets, fasteners, and/or welding.

[0072] The transom channel **806** of each transom housing **702** is disposed between the bottom coupling **802** and the aft coupling **804**. The transom channel **806** provides a smooth transition from the bottom coupling **802** to the aft coupling **804**. As shown in FIG. **9**, the aperture **712** of each transom housing **702** is axially aligned with the respective aft coupling **804**. As a result, the drive shafts **306** and the impellers **308** can pass through the apertures **712**, respectively, and, in turn, be automatically centered in the respective aft coupling **804** about an axis **960** that is coaxially aligned with the respective central axis **560**.

[0073] Turning now to FIGS. **10** and **11**, the transom plate **704** is provided to ensure proper alignment of the jet pump drive systems **202** and the transom assembly **204** with the bottom plate **206**, as shown in FIG. **2**. The transom plate **704** includes alignment apertures **1002**. In some examples, fasteners (not shown) can be disposed in the alignment apertures **1002** (and corresponding apertures formed in the bottom plate **206**) to align the transom plate **704** with the bottom plate **206** such that the transom plate **704** is aligned with the jet pump drive systems **202**. In turn, at least in this example, the transom plate **704** is disposed perpendicular to the bottom plate **206**. By auto-aligning the transom plate **704** with the other components of the jet pump assembly **200**, the jet pump assembly **200** is easier to assemble and can be assembled apart from the maritime vehicle (e.g., the maritime vehicle **100**).

[0074] The transom plate **704** includes impeller apertures **1004** and a steering aperture **1006**. The impeller apertures **1004** are sized and arranged to receive the impellers **308**, respectively. In the present example, the jet pump assembly **200** includes one steering assembly (not shown) configured to control the direction of both jet pump drives **202** via the single steering aperture **1006**. In other examples, the transom plate **704** can include more steering apertures **1006** to accommodate additional steering assemblies.

[0075] As shown in FIG. **10**A, the transom plate **704** includes a plurality of apertures **1020** for receiving various fasteners (e.g., apertures to secure the transom channel **702** to the transom plate **704**). In the present example, apertures **1020** are circumferentially arranged about each of the impeller apertures **1004**. Each of the plurality of apertures **1020** is a blind aperture (i.e., is not visible on the front surface **1052** or, more generally, from the front of the transom plate **704**, as shown in FIG. **10**B). As a result, none of the plurality of apertures **1020** can be a source of water leakage. Additionally, by reducing the amount of manufacturing time required to seal each of the plurality of apertures **1020**, the manufacturing time for the jet pump assembly **200** is reduced.

[0076] The transom plate **704** is coupled to the transom box **706** to form a watertight enclosure. In the present example, the front surface **1052** of the transom plate **704** includes slots **1054** that are disposed about the perimeter of the front surface **1052** and are configured to couple the transom plate **704** to the transom box **706**.

[0077] As illustrated in FIG. **11**, the transom box **706** includes a top wall **1202** and two side walls **1204** coupled to the top wall **1202**. Each of the top and side walls **1202**, **1204** includes tabs **1208** that are inserted into the slots **1054** disposed in the front surface **1102** of the transom plate **704** so as to couple the transom box **706** to the transom plate **704**. The transom box **706** also includes a lip **1212** that extends outward (upward in the orientation shown in FIG. **11**) from the top and side walls **1202**, **1204**. The lip **1212** can be sealed against an interior surface of the hull (e.g., the hull **104**) of the maritime vehicle. The lip **1212** can, for example, be welded, riveted, or otherwise fastened to the hull. As shown in FIG. **15**, the lip **1212** is fastened to the hull via bolts.

[0078] Finally, FIG. **12** illustrates the transom box **706** when coupled to the transom housings **702**, the transom plate **704**, and the bottom plate **206**. The transom box **706**, along with the transom plate **704** and the bottom plate **206**, form an enclosure open to the outside environment of the maritime vehicle **100**. In the present example, at least part of the impeller **308** and the nozzle **602** are disposed in the transom box **706** (and the enclosure defined by the transom box **706**).

Bottom Plate

[0079] FIGS. **13**A & **13**B illustrate the bottom plate **206** in greater detail. The bottom plate **206** is configured to secure and align the jet pump drive system **202** and the transom assembly **204**. In the present example, the bottom plate **206** is a single piece of metal that includes a base **1300**, a plurality of apertures **1302**, and a plurality of pockets. The apertures **1302** are disposed in the base **1300** to help align and secure the components of the jet pump drive system **202** and the transom assembly **204**, whereas the pockets are sized and arranged in the base **1300** to receive various components of the jet pump drive system **202** and the transom assembly **204**. In other examples, the bottom plate **206** can be made of two or more sheets of metal fastened or welded together.

[0080] In the present example, the bottom plate **206** includes five pockets—pockets **1312**, **1314**, and **1316**—formed in the base **1300**. The first and second pockets **1312** of the bottom plate **206** are sized to receive and retain the motors **302**. The third and fourth pockets **1314** include openings **1315** that extend through the bottom plate **206** and are sized to receive and be enclosed by the bottom coupling **802** of the transom housings **702**, respectively, when the coupling plates **822** of the transom housing **702** are seated against the portions of the pockets **1314** surrounding the openings **1315**, respectively. Finally, the fifth pocket **1316** is sized to receive the transom plate **704** and the transom box **706**.

[0081] As shown in FIGS. **13**A, **13**B, and **14**, the base **1300** of the bottom plate **206** has an upper surface **1400** and a lower surface **1402** opposite the upper surface **1400**. As best shown in FIG. **14**, the bottom plate **206** also includes a lip **1404** offset from the lower surface **1402** of the base **1300** by a distance **1406**. The lip **1404** is offset from the lower surface **1402** so that the lower surface **1402** can be flush with an external surface of the hull of the maritime vehicle. As a result, the distance **1406** is approximately equal to the thickness of the hull. In turn, when the lip **1404** is coupled to an interior surface of the hull, the lower surface **1402** can help create a flush, uniform exterior surface of the maritime vehicle. By creating a flush, uniform exterior surface, the lower surface **1402** reduces drag and reduces the likelihood of generating cavitation along the exterior surface of the hull.

[0082] The lip **1404** generally extends along three of the four sides of the bottom plate **206**. In some examples, the bottom plate **206** also includes a vertical lip **1408** that extends from the lip **1404**. The vertical lip **1408** is configured to abut the lip **1212** of the transom box **706**. Preferably, the lip **1408** and the lip **1212** can be sealed to one another with welding and/or marine-grade sealant to make a watertight seal between the transom box **706** and the bottom plate **206**. In other examples, a seal can be provided between the lip **1212** and the lip **1404**.

Assembly and installation

[0083] In FIG. **2**, the dual jet pump assembly **200** is fully assembled, and FIG. **15** illustrates the fully assembled dual jet pump assembly **200** installed in the maritime vehicle **100**. Beneficially, the dual jet pump assembly **200** can be fully assembled apart from the maritime vehicle **100**. In other examples, the dual jet pump assembly **200** can be simultaneously assembled and installed on the maritime vehicle **100**. When assembling the dual jet pump assembly **200**, either the motors **302** or the transom assembly **204** can be installed on the bottom plate **206** first. After the transom assembly **204** and the motors **302** are installed on the bottom plate **206**, the rest of the jet pump drive system **202** can be installed.

[0084] The transom assembly **204** is coupled to the bottom plate **206** by first coupling the transom plate **704** to the bottom plate **206**. As discussed above, the transom plate **704** includes the alignment apertures **1002** to ensure the transom plate **704** is properly aligned with the bottom plate **206**, and fasteners (e.g., pins, not shown) can be disposed in the alignment apertures **1002** and alignment apertures **1322** formed in the bottom plate **206** (see FIG. **13**B) and aligned with the alignment apertures **1002** when the transom plate **704** is properly aligned with the bottom plate **206**. Next, the transom box **706** is coupled to the transom plate **704** and the bottom plate **206**. In various examples, the lip **1212** of the transom box is aligned with the lip **1404** of the bottom plate **206** and the edge of the lip **1212** is sealed to the edge of the lip **1404**. The transom housings **702** are then coupled to the bottom plate **206** and the transom plate **704**.

[0085] To couple the motors **302** to the bottom plate **206**, motor brackets **212** are installed on the bottom plate **206** (see FIG. **15**). In some examples, the brackets **212** include alignment pins (not shown) to ensure the brackets **212** are properly placed and aligned on the bottom plate **206**. The motors **302** can then be secured in the motor brackets **212**.

[0086] With the motors **302** and the transom assembly **204** installed on the bottom plate **206**, the drive shafts **306**, the impellers **308**, and the nozzles **602** can be installed. In the present example, the first end **502** of each drive shaft **306** is passed through the transom assembly **204**, through the aperture **712** of a respective one of the transom housings **704**, and coupled to a respective one of the motors **302** via the respective coupling **312**.

[0087] At this point, the jet pump assembly **200** is functionally assembled. The jet pump assembly **200** can then be sealed so as to be watertight. In various examples, joints between the bottom plate **206** and the transom assembly **204** can be welded or sealed with marine-grade sealants. With the components of the jet pump assembly **200** functionally assembled and fully sealed, the jet pump assembly **200** can be bench tested and installed on the vehicle.

[0088] As shown in FIGS. **15** and **16**, the jet pump assembly **200** is bolted to the hull **104** of the maritime vehicle **100**. In other examples, the jet pump assembly **200** can be rivetted or welded to the hull **104**. Installing the jet pump assembly **200** can be done entirely from the inside of the maritime vehicle **100**. Unlike other jet pump assemblies, the jet pump assembly **200** does not require any underside access to the maritime vehicle **100**. Furthermore, because the jet pump assembly **200** is already functional and properly assembled, the jet pump assembly **200** does not require extensive alignment and testing after installed in the maritime vehicle **100**. Additionally, in some examples, the weight of the jet pump assembly **200** is low enough that the jet pump assembly **200** can be fully assembled, tested, and installed by a single person.

[0089] The jet pump assembly **200** is lowered into the opening at the aft, bottom portion of the maritime vehicle **100**. The lip **1212** and the lip **1404** are inserted against the interior surface of the hull **104**. In some examples, the lips **1212**, **1404** are riveted, bolted (or otherwise fastened), or welded to the hull **104**. In some examples, the jet pump assembly **200** further requires marine sealant to prevent or reduce water leaks between the jet pump assembly **200** and the hull **104** of the maritime vehicle **100**.

[0090] After installing the jet pump assembly **200** to the maritime vehicle **100**, the jet pump assembly **200** can be coupled to the various other systems and components of the maritime vehicle. For example, the motors **302** can be electrically connected to a power source (not shown) and the electronic controller(s) (not shown). In some examples, the power source is a battery, generator, or other electrical power source. Additionally, the jet pump assembly system **200** can be functionally connected to a vehicle cooling system (e.g., cooling system **1700** described in greater detail in connection with FIG. **17**).

Operation of the Jet Pump Assembly

[0091] During operation, water first enters the water intake of the maritime vehicle **100** and proceeds through the transom channels **806** to the impellers **308**. The water intake generally draws water into the transom channels **806** through an opening or grate in the bottom of the maritime vehicle **100**, e.g., the openings **1315**. In some examples, the openings **1315** and transom channels **806** are fully submerged when the maritime vehicle **100** is floating in a body of water.

[0092] The impellers **308** are disposed on the drive shafts **306**, respectively, and at least partly disposed in the transom channels **806**, respectively. As discussed above, each of the rotating impellers **308** typically includes several blades **552**. In some examples, the blades **552** are shaped (e.g., curved) to improve the hydrodynamic efficiency of the impeller **308**. As the impellers **308** spin, the impellers **308** impart momentum to the water, increasing its velocity.

[0093] The water, after passing through the respective impeller **308**, is directed into the respective diffuser **604**. The respective diffuser **604** gradually expands the flow area, converting the kinetic energy of the high-speed water into pressure energy. The pressurized water exiting the respective diffuser **604** is then directed through the specially shaped, respective nozzle **606** at the rear of the jet pump assembly **200**. As the pressurized water exits the respective nozzle **606**, the water creates a reactive force in the opposite direction, propelling the maritime vehicle **100** forward through the water. The respective nozzle **606** also includes the directional outlet **608** that can be tilted to control the direction of the water jet. By adjusting the angle of the directional outlets **608**, the operator can steer the maritime vehicle **100**.

Operating Efficiency of the Jet Pump Assembly

[0094] In some examples, the jet pump assembly **200** provides enough propulsive thrust for the maritime vehicle **100** to transition from a displacement mode or regime (e.g., the vehicle is mostly submerged and pushes water aside during movement) to a planning mode or regime (e.g., the vehicle lifts partially or mostly out of the water and may skim across the surface of the water). The hump speed or transition speed is when the maritime vehicle **100** transitions from the displacement mode to the planing mode. Below the hump speed, the vehicle **100** typically operates in the displacement mode, but above the hump speed, the vehicle **100** transitions into the planing mode. Conventionally, the efficiency of jet-based pump systems is proportional to the speed of the vehicle relative to the water, such that jet-based pump systems are typically less efficient in a displacement mode than in a planning mode. However, the jet pump assembly **200** described herein is maximally efficient in both a displacement and a planing regime.

Cooling System

[0095] FIG. **17** is a schematic diagram of an example cooling system **1700** that can be employed in the maritime vehicle **100** to reduce or maintain the temperature of various components of the jet pump assembly **200**, particularly the motors **302** and the motor controllers for the motors (not shown). The cooling system **1700** generally includes a pump **1702**, a reservoir **1704**, and one or more heat exchangers **1706**. In some examples, the cooling system **1700** includes a single pump **1702** and a single reservoir **1704** fluidly coupled to one heat exchanger **1706**. In other examples, however, the cooling system **1700** can include one pump **1702** and one reservoir **1704** fluidly coupled to two or more heat exchangers **1706**, as illustrated in dashed lines in FIG. **17**. Further, while the cooling system **1700** is described in connection with the jet pump assembly **200**, it will be appreciated that the cooling system **1700** can instead be used to cool the components of a different type of thrust system other than the jet pump assembly **200**.

[0096] In the present example, the pump **1702** pumps a coolant through an enclosed piping system **1708** that fluidly connects the pump **1702**, the reservoir **1704**, the heat exchanger(s) **1706**, and all other components of the cooling system **1700**. The piping system **1708** can include piping and various couplings and junctions (not shown) that enable the pump **1702** to pump coolant through the cooling system **1700**. For example, the enclosed piping system **1708** may include fittings to couple with couplings **422**, **424** of the cooling jacket **412** of each of the motors **302**, as described above. The enclosed piping system **1708** is preferably made of one or more materials that will not corrode or fail at the operating temperatures of the coolant and the cooling system **1700**. It will be appreciated that the coolant can include any known coolant. For example, the coolant may be a water and glycol solution.

[0097] In operation, the pump **1702** pumps the coolant from the reservoir **1704** through the jet pump assembly **200** so as to cool the electronic components of the jet pump assembly **200** as needed. More particularly, the pump **1702** pumps the coolant through, for example, the motor cooling jacket **412** of each of the motors **302** and/or a heat sink **1714** for each of the motor controllers (e.g., the electronic speed controller(s)). In some examples, the enclosed piping system **1708** can include controllable valves (not shown) to control how much coolant is distributed between the motor cooling jacket(s) **412** and the heat sink(s) **1714**. In some examples, the cooling system **1700** can be utilized to cool other components of the maritime vehicle **100** (e.g., a power source, other computer systems).

[0098] Heat generated by the motor(s) **302** and the motor controller(s) is transferred to the coolant passing through the motor cooling jacket(s) **412** and/or the heat sink(s) **1714**. As a result, the coolant is hotter after passing through the motor cooling jacket **412** and/or the heat sink(s) **1714**. The hot(ter) coolant then passes to the heat exchanger(s) **1706**. In some examples, the enclosed piping system **1708** includes a coupling **1722** before the heat exchanger(s) **1706** and a coupling **1724** after the heat exchanger(s) **1706**. The couplings **1722**, **1724** will be described in greater detail below.

[0099] Each heat exchanger **1706** includes an inlet **1728** that receives the hot(ter) coolant and an outlet **1732** at which chilled or cooled coolant is provided. In the present example, the heat exchanger **1706** can be a micro-keel cooler such as the micro-keel cooler described in greater detail in connection with FIGS. **18-22**. In any event, during operation of the heat exchanger(s) **1706**, heat is transferred away from the hot(ter) coolant through the heat exchanger(s) **1706**. As described in greater detail below, the heat exchanger(s) **1706** convey(s) the heat transferred away from the coolant into the water surrounding the maritime vehicle **100**.

[0100] In some examples, the couplings **1722**, **1724** are used to couple sensors to the cooling system **1700** to detect characteristics of the coolant. For example, one or both of the couplings **1722**, **1724** can include a pressure sensor and/or a temperature sensor. In some examples, the sensors send sensor signals to a controller that manages the operation of the pump **1702** to ensure the heat exchanger(s) **1706** operate(s) within acceptable ranges and, preferably, optimized temperature ranges.

[0101] Finally, the cooled coolant passes from the outlet **1732** of the heat exchanger(s) **1706** back to the reservoir **1704**. The reservoir **1704** is provided to accommodate volume expansion of coolant undergoing large temperature changes. The reservoir **1704** may also store extra coolant. In some examples, the reservoir **1704** can store an extra 33 percent (%) to 66% of the volume of the coolant flowing through the cooling system **1700**.

Micro-Keel Cooler

[0102] FIGS. **18-22** illustrate an example micro-keel cooler **1800** made in accordance with the present disclosure. In the present example, the micro-keel cooler **1800** is shown in use with the jet pump assembly **200**. In particular, the micro-keel cooler **1800** is disposed in a recess **1801** (shown in FIG. **15**) adjacent to the jet pump assembly **200** installed on the maritime vehicle **100**, such that the micro-keel cooler **1800** is configured to cool the components of the jet pump assembly **200** when employed in the maritime vehicle **100**. In other examples, however, the micro-keel cooler **1800** can be used in connection with a different type of thrust system (e.g., a different jet-based pump thrust system) and/or a different maritime vehicle other than the maritime vehicle **100**.

[0103] With reference first to FIG. **18**, the micro-keel cooler **1800** generally includes an inlet **1802**, an outlet **1804**, and means for reducing a temperature of fluid flowing from the inlet **1802** to the outlet **1804**. In the present example, the inlet **1802** and the outlet **1804** are identical such that the micro-keel cooler **1800** is symmetrical. In other examples, the micro-keel cooler **1800** can be asymmetrical. Generally speaking, the means for reducing the temperature of fluid flowing from the inlet **1802** to the outlet **1804** should be designed to maximize turbulent flow. In the present example, the means takes the form of cylindrical tubes **1806** that are disposed between the inlet **1802** and the outlet **1804**. In other examples, however, the means can take the form of tubes, hoses, pipes, or other fluid conveying objects having a different shape (e.g., the means may include sharp internal corners of a triangle or a square instead of an internal circular shape).

[0104] As illustrated in FIG. **18**, the inlet **1802** of the micro-keel cooler **1800** includes an inlet connection **1822** and an inlet mixing chamber **1824**. When the heat exchanger **1706** discussed above takes the form of the micro-keel cooler **1800**, the inlet connection **1822** receives the hot(ter) coolant after the coolant has passed through the motor cooling jacket **412** and/or the heat sink **1714**. The inlet mixing chamber **1824** is at least partially enclosed by an inlet endcap **1826**. The inlet mixing chamber **1824** ensures that the coolant received via the inlet connection **1822** is properly mixed before passing through the tubes **1806** to the outlet **1804**. The outlet **1804** includes an outlet connection **1832** and an outlet mixing chamber **1834**. When the heat exchanger **1706** takes the form of the micro-keel cooler **1800**, the outlet connection **1832** is arranged to provide chilled or cooled coolant to the reservoir **1704**. In the present example, the outlet mixing chamber **1834** is identical to the inlet mixing chamber **1824**, such that the outlet mixing chamber **1834** is also at least partially enclosed by an outlet endcap **1826**. It will be appreciated that the tubes **1806** extending between the inlet **1802** and the outlet **1806**, the spacing of the tubes **1806**, and the total surface area of the micro-keel cooler **1800** facilitate the transfer of heat from the coolant as the coolant flows from the inlet mixing chamber **1824** to the outlet mixing chamber **1834** and out of the system via the outlet **1804**.

[0105] As shown in FIG. **19**, the micro-keel cooler **1800** may further include an inlet bracket **1902**, an outlet bracket **1904**, and one or more plates **1906** coupled to each of the inlet bracket **1902** and the outlet bracket **1904**. The inlet bracket **1902** is arranged immediately adjacent the inlet **1802** (but on the opposite surface of the micro-keel cooler **1800** from the inlet **1802**). Likewise, the outlet bracket **1904** is arranged immediately adjacent the outlet **1804** (but on the opposite surface of the micro-keel cooler **1800** from the outlet **1804**). The plates **1906** extend outward (downward in FIG. **18**B) from the plurality of tubes **1806** such that the plates **1906** are positioned to direct the water surrounding the maritime vehicle **100** over and between the tubes **1806**, which helps to further remove heat from the coolant flowing through the tubes **1806**. In the present example, the micro-keel cooler **1800** includes three plates **1906**, and each plate **1906** extends along the entire length of the micro-keel cooler **1800** between the inlet bracket **1902** and the outlet bracket **1904**. In other examples, however, the micro-keel cooler **1800** can include more or less plates **1906** and/or the plates **1906** can extend along only part of the length of the micro-keel cooler **1800**. In some examples, the plates **1906** can include or be coupled to one or more hydrodynamic surfaces that help to direct the surrounding water over one or more of the tubes **1806**. For example, the hydrodynamic surfaces may comprise an airfoil that directs the surrounding water between the tubes **1806**. The hydrodynamic surfaces can improve the efficiency of the micro-keel cooler **1800**. In some examples, the plates **1906** are simply snapped into the inlet and outlet brackets **1902**, **1904**. In other examples, the plates **1906** are fastened or welded to the inlet and outlet brackets **1902**, **1904**.

[0106] With reference now to FIGS. **19-21**, the micro-keel cooler **1800** is assembled by coupling together a bottom plate **2000** including a first tube array **2002** (shown in FIG. **20**), a top plate **2100** including a second tube array **2102** (shown in FIG. **21**), and the inlet and outlet endcaps **1826** (shown in greater detail in FIG. **22**). In the present example, the first tube array **2002** includes nine tubes **2002** and the second tube array **2102** includes eight tubes **2102**. In other examples, the first and second tube arrays **2002**, **2102** can include more or fewer tubes.

[0107] As best illustrated in FIG. **20**, the bottom plate **2000** includes a first half **2012** of the inlet mixing chamber **1824**, a first half **2014** of the outlet mixing chamber **1834**, and the first tube array **2002**. As best illustrated in FIG. **21**, the top plate **2100** includes a corresponding second half **2112** of the inlet mixing chamber **1824**, a corresponding second half **2114** of the outlet mixing chamber **1834**, and the second tube array **2102**. The first tube array **2002** is first welded or otherwise secured to the first halves **2012**, **2014**. The second tube array **2102** is similarly welded or fastened to the corresponding halves **2112**, **2114**.

[0108] When the bottom plate **2000** and the top plate **2100** are coupled together, the halves **2012**, **2014**, **2112**, **2114** of the inlet and outlet mixing chambers **1824**, **1834** are coupled together. In the present example, the halves **2012**, **2112** and the halves **2014**, **2114** are coupled together via welding, adhesives, fasteners, sealants, or any combination thereof. Likewise, the first and second tube arrays **2002**, **2102** are coupled together to form the tubes **1806**.

[0109] The inlet mixing chamber **1824** and the outlet mixing chamber **1834** are capped, or enclosed, with the inlet and outlet endcaps **1826**, respectively. As shown in FIG. **22**, each endcap **1826** includes a plug **2202** disposed on an end plate **2204**. The plug **2202** includes a sidewall **2206** that extends outward from the end plate **2204**. In the present example, the plug **2202** of each endcap **1826** is inserted into the respective mixing chamber **1824**, **1834** and the sidewall **2206** of that endcap **1826** sealingly engages an internal surface of the micro-keel cooler **1800** defining the inlet mixing chamber **1824** or the outlet mixing chamber **1834**. In some examples, the plug **2202** can further include welding or marine sealant to ensure the mixing chambers **1824**, **1834** are watertight.

[0110] The micro-keel cooler **1800** is assembled in the foregoing manner because the micro-keel cooler **1800** is too small to assemble in a more typical manner. For example, the tubes **1806** cannot be welded or fastened to the inlet and outlet mixing chambers **1824**, **1834** if the inlet and outlet mixing chambers are not manufactured in subsequently coupled halves as described herein.

Micro-Keel Cooler Cooling Capacity

[0111] In the present example, the maritime vehicle **100** includes two micro-keel coolers **1800**, as best illustrated in FIGS. **16** and **23**. In the present example, the two micro-keel coolers **1800** are located on opposing sides of the keel **124**, with one of the micro-keel coolers **1800** immediately adjacent one of the sides **120** and the other of the micro-keel coolers **1800** immediately adjacent the other side **120** and separated by the jet pump assembly **200**. In this manner, one of the micro-keel coolers **1800** can cool the components of one of the jet pump drive systems **202** (more particularly the motor **302** of that jet pump assembly **202**), whereas the other micro-keel cooler **1800** can cool the components of the other jet pump assembly **202** (more particularly the motor **302** of that jet pump assembly **202**).

[0112] In some examples, the micro-keel coolers **1800** are capable of providing the total cooling capacity of the vehicle. In other examples, the micro-keel coolers **1800** supplement additional cooling systems (e.g., cooling exhaust systems) carried by the maritime vehicle **100**. In various examples, each micro-keel cooler **1800** is able to transmit approximately 1500 Watts to 2500 Watts of heat energy into the surrounding water. However, in some examples, each micro-keel cooler **1800** can improve heat transfer capacity and transfer as much as approximately 3500 Watts or 4500 Watts of heat energy into the surrounding water.

[0113] The micro-keel cooler **1800** is designed for improving heat dissipation while being small in size. In the present example, each of the tubes **1800** can have a diameter between approximately 0.25 inches (in.) diameter to 0.75 in. Additionally, each of the tubes **1800** can be between approximately 6 in. and 36 in. in length. While the micro-keel cooler **1800** includes 17 tubes, in some examples, the micro-keel cooler **1800** can include more or fewer tubes. Moreover, while in the present example the micro-keel cooler **1800** includes two layers of tubes **1806**, in other examples, the micro-keel cooler **1800** can include one or more than two layers of tubes **1806**. Furthermore, while in the present example, the distance between the center of adjacent tubes **1806** is preferably between approximately 0.3 in. and approximately 1 in, it will be appreciated that the spacing between the tubes **1806** can be adjusted.

[0114] Additionally, the micro-keel cooler **1800** can further adjust the amount of heat dissipated into the surrounding water based on the coolant, coolant flow, and the conditions of the surrounding water. The coolant can be selected for an improved specific heat capacity. The coolant can, for example, have a specific heat capacity of between approximately 2 kilojoules per kilogram degree Kelvin (kJ/kgK) and approximately 5 kJ/kgK. Additionally, the coolant system **1700** is configured to pump between approximately 0.5 gallons per minute (gal/min) and approximately 3 gal/min of coolant through each micro-keel cooler **1800**. Further, the ambient environment of the vehicle can fluctuate between approximately −5 degrees Celsius (° C.) to approximately 40° C.

[0115] The dual jet pump assembly **200** and the cooling system **1700** incorporating the micro-keel cooler **1800**, as described herein, provide several benefits over other maritime vehicle power and cooling systems.

[0116] First, the dual jet pump assembly **200** can be operated with improved efficiency points. For example, the dual jet pump assembly **200** can operate efficiently in both a displacement regime and a planing regime. Furthermore, the dual jet pump assembly **200** can transition the vehicle from a displacement regime to a planing regime more efficiently than other systems. As a result, the dual jet pump assembly **200** causes the maritime vehicle **100** to be more versatile than other maritime vehicles.

[0117] Second, the dual jet pump assembly **200** is designed to be assembled as a lightweight, functional unit. For example, the dual jet pump assembly **200** can be manufactured and assembled prior to installation in a vehicle. This simplifies the installation process. Furthermore, the lightweight nature of the dual jet pump assembly **200** allows a single person to handle and install the dual jet pump assembly **200** in the maritime vehicle **100**.

[0118] Third, the dual jet pump assembly **200** can be installed as an operational unit. As a result, a technician installing the dual jet pump assembly **200** does not need to have access to an underside of the maritime vehicle **100**. The installation of the jet pump assembly **200** can be completed from an interior of the maritime vehicle **100**. Further, maintenance and replacement are simpler than other jet pump systems because maintenance can also be conducted from the internal cavity of the maritime vehicle **100**.

[0119] Fourth, the cooling system **1700** utilizes the micro-keel cooler **1800** to dissipate the heat generated by the dual jet pump assembly **200** into the environment (e.g., the water) surrounding the maritime vehicle **100**. The small size of the micro-keel cooler **1800** provides extensive cooling in a package or profile smaller than typical heat exchangers, which is advantageous given the small nature of the maritime vehicle **100**.

[0120] Those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above-described examples without departing from the spirit and scope of the invention(s) disclosed herein, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept(s).

[0121] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

Claims

1. A jet pump assembly for causing a maritime vehicle to traverse a body of water, the jet pump assembly comprising: a jet pump drive system including an impeller and a motor operatively coupled to the impeller to drive rotation of the impeller; a transom assembly including: a transom plate including a first aperture facing the motor; a transom housing coupled to the transom plate, thereby enclosing the first aperture; and a transom box coupled to the transom plate and including a second aperture aligned with the first aperture; and a bottom plate coupled to and disposed perpendicular to the transom plate, the bottom plate including: a base; and an opening formed through the base, wherein the opening is enclosed by the transom housing, wherein the impeller is at least partially disposed within the transom box.

2. The jet pump assembly of claim 1, wherein the transom housing further includes a third aperture aligned with the first aperture of the transom plate, wherein the jet pump drive system further includes a drive shaft that operatively couples the motor to the impeller, and wherein at least part of the drive shaft is disposed in the transom channel and passes through the aperture.

3. The jet pump assembly of claim 1, wherein the impeller includes an impeller head and a plurality of impeller blades disposed about the impeller head, wherein the impeller blades are rotatable about a central axis, and wherein the drive shaft is centered about an axis that is coaxially aligned with the central axis.

4. The jet pump assembly of claim 1, wherein the jet pump drive system further comprises a directional nozzle coupled to the impeller.

5. The jet pump assembly of claim 4, wherein the directional nozzle is at least partially disposed within the transom box.

6. The jet pump assembly of claim 1, wherein the transom plate is coupled to the bottom plate via alignment fasteners inserted into alignment apertures formed in the transom plate and corresponding alignment apertures formed in the bottom plate.

7. A method of assembling a jet pump assembly, comprising: assembling a transom assembly, including: coupling a transom plate, having at least a first aperture, to a transom box, having a corresponding second aperture; and coupling a transom housing, having an inlet and an outlet, to the first aperture of the transom plate, such that the transom housing encloses the first aperture of the transom plate and the outlet of the transom housing is fluidly coupled to the first aperture of the transom plate; installing a jet pump drive system on the transom assembly, the jet pump drive system including an impeller and a motor operatively coupled to the impeller to drive rotation of the impeller; coupling the transom plate to a bottom plate, the bottom plate including an opening extending through the bottom plate; and enclosing the opening with the transom channel, such that the opening is fluidly coupled to the inlet of the transom channel.

8. The method of claim 7, wherein installing the jet pump drive system on the transom assembly includes: inserting a drive shaft, having a first end and a second end, through the first aperture of the transom plate and the second aperture of the transom box and through a third aperture in the transom housing; coupling the motor to the first end of the drive shaft; coupling the impeller to the second end of the drive shaft.

9. The method of claim 8, further comprising coupling the jet pump drive system to the bottom plate.

10. The method of claim 9, wherein coupling the jet pump drive system to the bottom plate includes: coupling the motor to a bracket, wherein the drive shaft passes through the bracket; inserting alignment pins on the bracket in corresponding alignment holes on the bottom plate; and fastening the bracket to the bottom plate.

11. The method of claim 7, wherein coupling the transom plate to the bottom plate includes inserting alignment pins on the transom plate into corresponding alignment holes on the bottom plate.

12. The method of claim 7, further comprising creating water-tight sealing joints between the transom assembly, the jet pump drive system, and the bottom plate.

13. A maritime vehicle configured to traverse a body of water, the maritime vehicle comprising: a hull; and a jet pump assembly, comprising: a jet pump drive system including an impeller and a motor operatively coupled to the impeller to drive rotation of the impeller; a transom assembly including: a transom plate including a first aperture facing the motor; a transom housing coupled to the transom plate, thereby enclosing the first aperture; and a transom box coupled to the transom plate and including a second aperture aligned with the first aperture and a first lip disposed along at least a portion of a perimeter of the transom box, wherein the first lip is configured to sealingly engage an interior surface of the hull; and a bottom plate coupled to and disposed perpendicular to the transom plate, the bottom plate including: a base; and an opening formed through the base, wherein the opening is enclosed by the transom housing.

14. The maritime vehicle of claim 13, wherein the impeller is at least partially disposed within the transom box.

15. The maritime vehicle of claim 13, wherein the base includes a lower surface wherein the bottom plate further includes a second lip that is offset from the lower surface and is disposed along at least one side of the bottom plate, and wherein the lower surface is flush with an exterior surface of the hull when the second lip is coupled to the interior surface of the hull.

16. A method of installing a jet pump assembly on a maritime vehicle, comprising: providing the jet pump assembly, the jet pump assembly including: a jet pump drive system including an impeller and a motor operatively coupled to the impeller to drive the impeller; a transom assembly including a transom plate, a transom box, and a transom housing; and a bottom plate including a base, an opening formed through the base, and an outer lip disposed along at least a portion of a perimeter of the bottom plate and offset from a lower surface of the base; coupling the outer lip of the bottom plate to an interior surface of a hull of the maritime vehicle; coupling the transom box to the interior surface of the hull of the vehicle; and installing the jet pump drive system in the transom assembly.

17. The method of claim 16, wherein coupling the outer lip of the bottom plate to an interior surface of a hull of the vehicle includes inserting the bottom plate in an opening disposed in the hull of the maritime vehicle.

18. The method of claim 16, further comprising aligning a lower surface of the bottom plate with an exterior surface of the hull.

19. The method of claim 16, wherein the jet pump drive system is installed in the transom assembly before coupling the bottom plate or the transom box to the interior surface of the hull of the vehicle.

20. The method of claim 16, wherein the transom assembly is coupled to the bottom plate before coupling the bottom plate to the interior surface of the hull of the vehicle.